



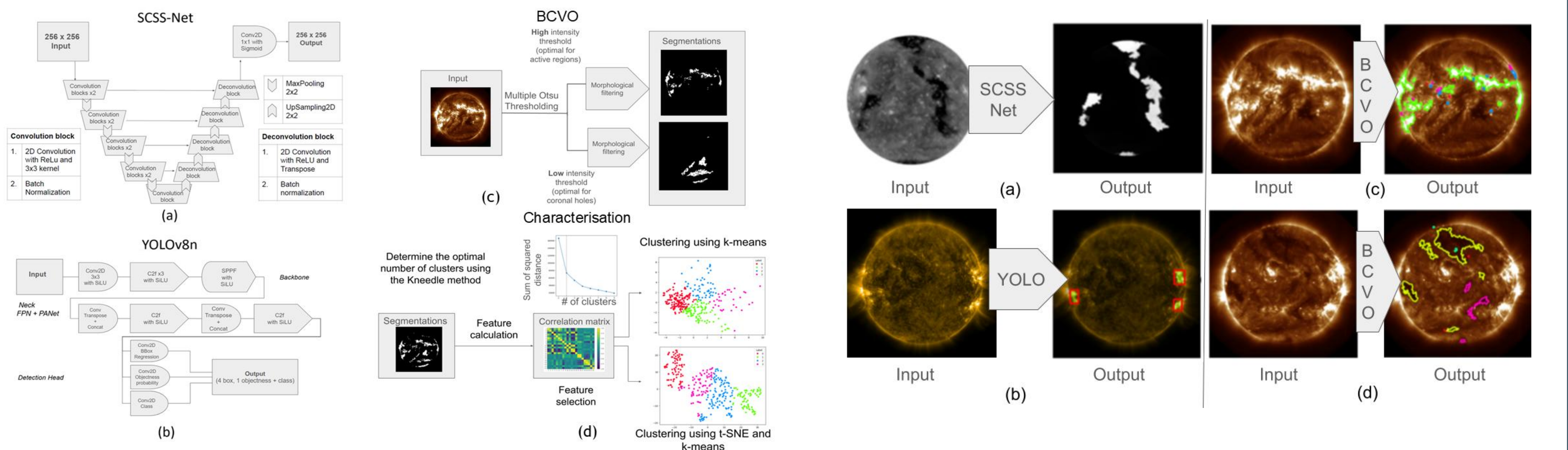
Solar Structure Segmentation for Onboard AI and Solar Science

Panagiotis Gonidakis, Centre for Plasma Astrophysics (CmPA), KULeuven, Belgium

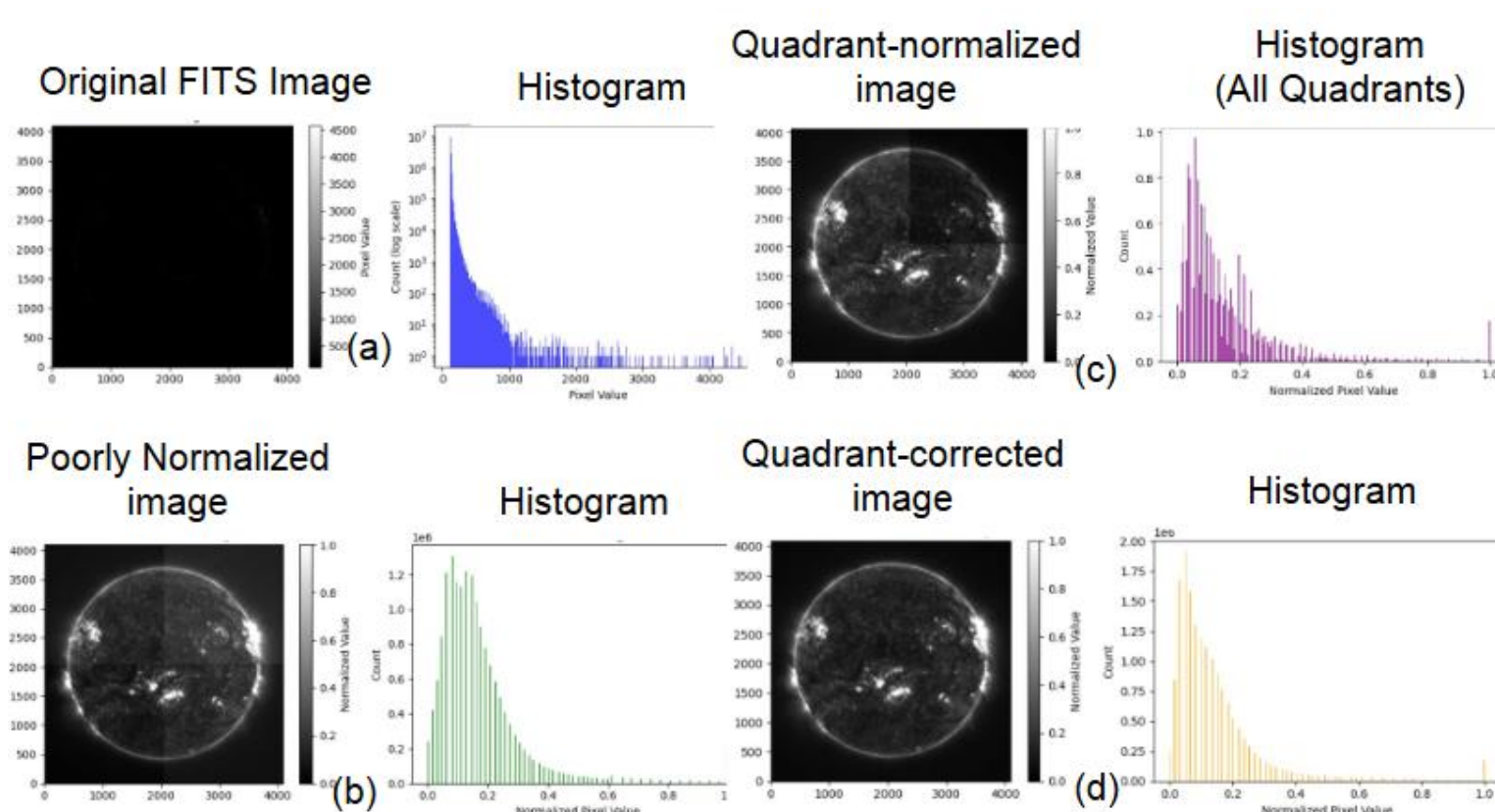
INTRODUCTION

Solar physics and **space weather forecasting** increasingly rely on the automated identification of coronal structures using machine learning. We assess the **feasibility of onboard deployment** by comparing **three** approaches. Each method is first trained and evaluated on well-calibrated SDO observations and subsequently tested on **unseen raw SDO data** to assess its robustness under conditions relevant to onboard operation.

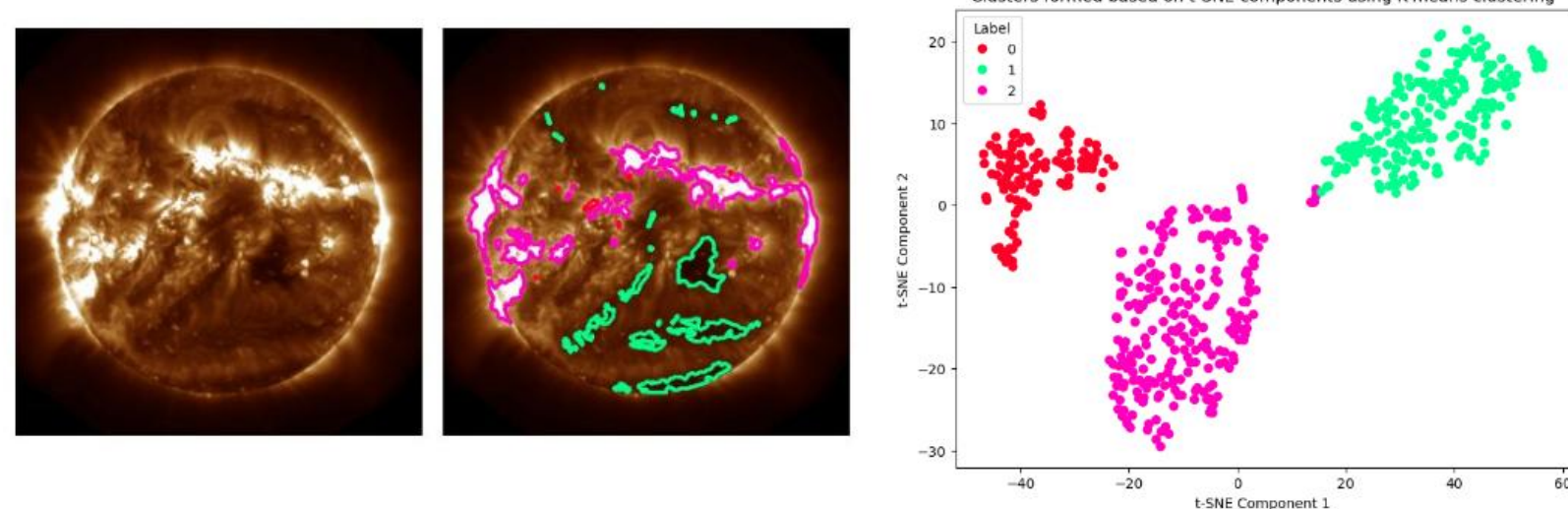
METHODS



RAW SDO DATA (Level - 0)



CHARACTERIZATION

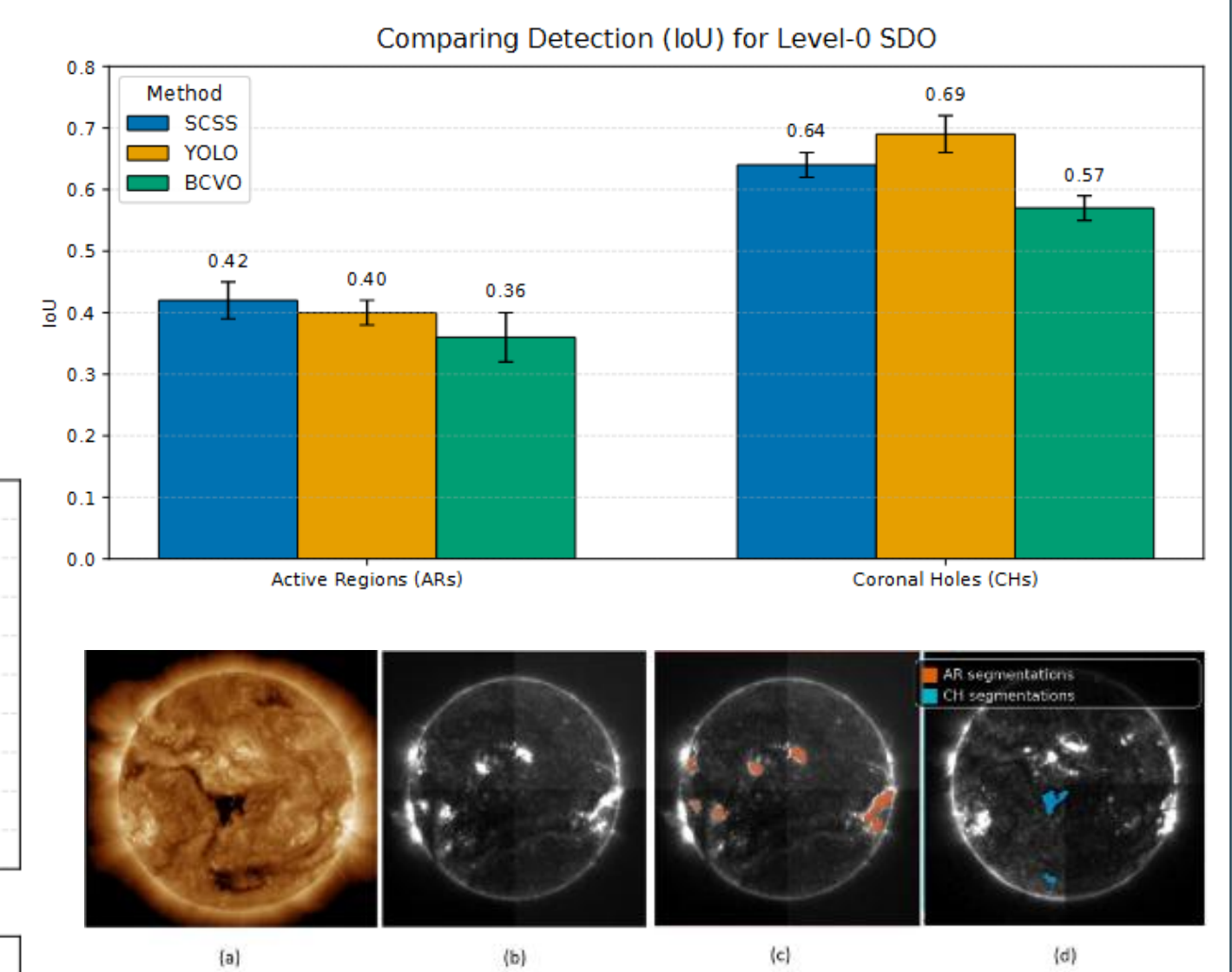
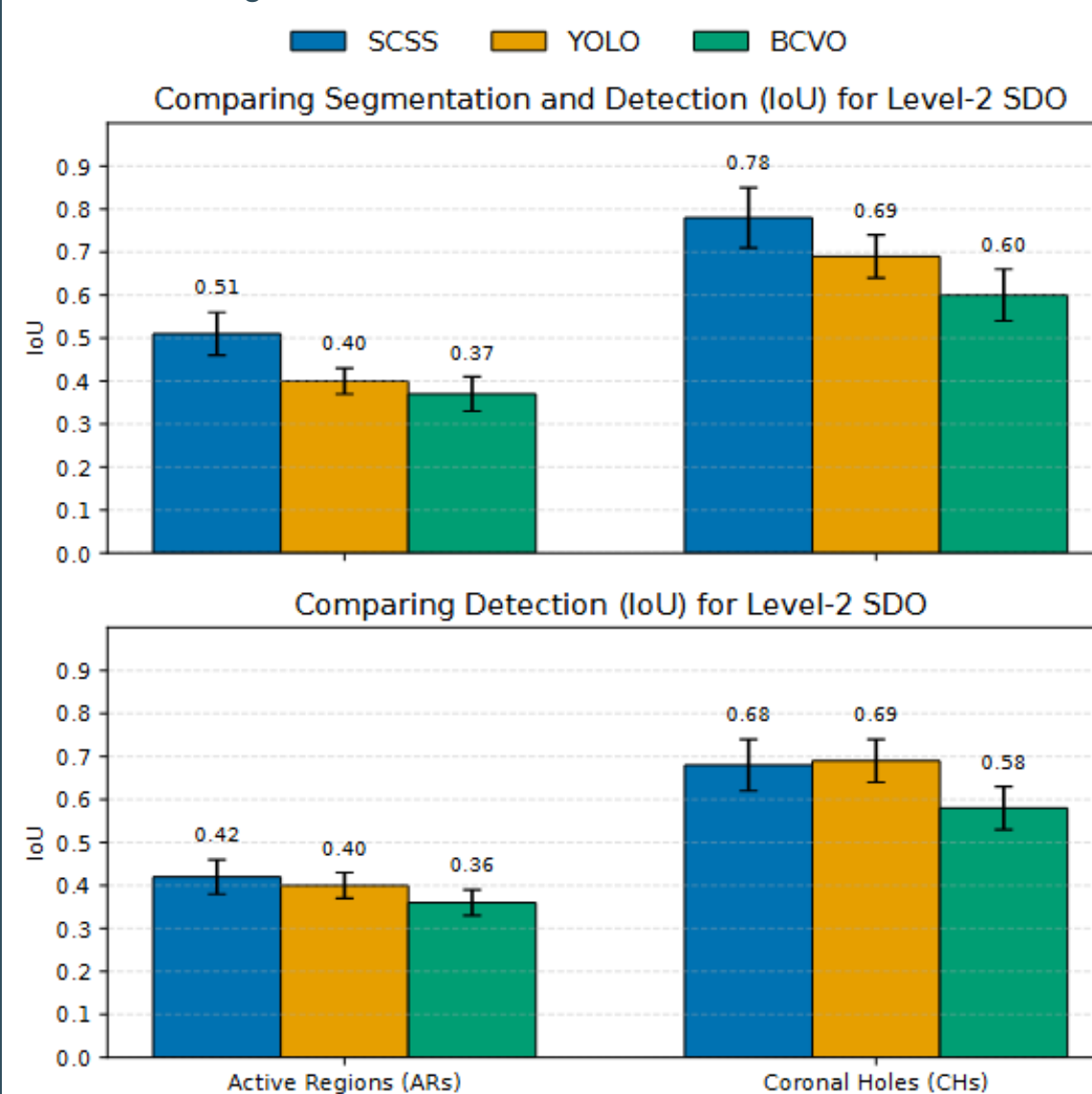


RESULTS

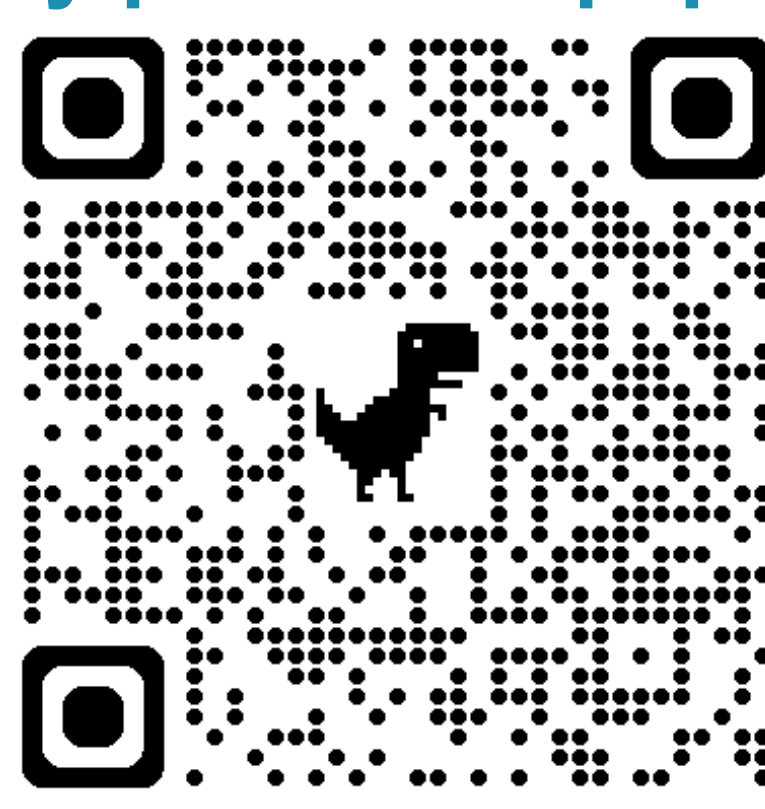
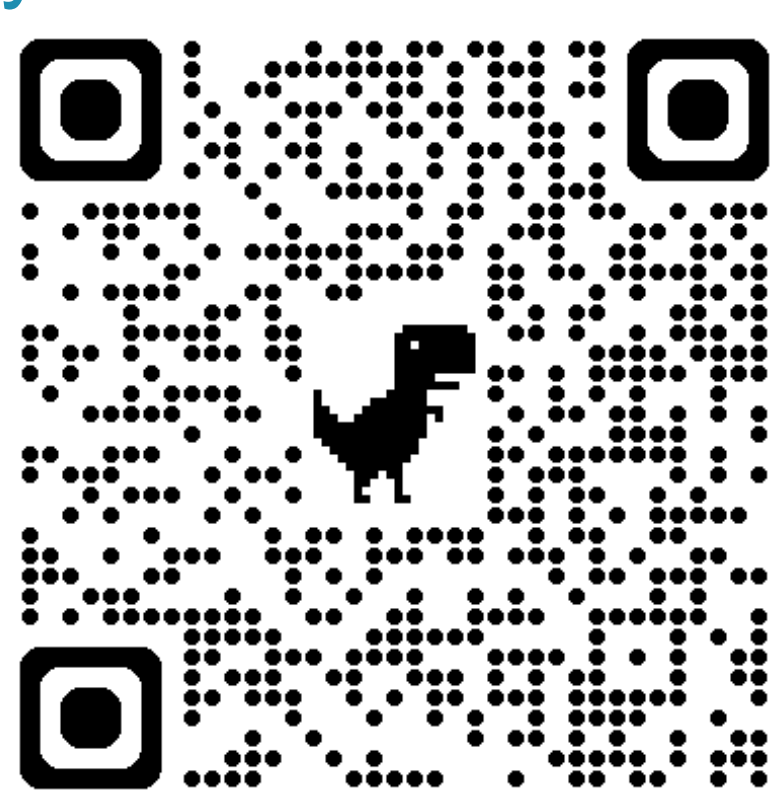
Intersection over Union (IoU)

$$IoU = \frac{|A \cap B|}{|A \cup B|}$$

where A is the predicted set of pixels (or bounding box) and B is the ground truth



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Comparing Solar Structure Detection Methods in SDO/AIA Observations and the Application to Raw Uncalibrated Data

P. Gonidakis, F. Carella, E. Dineva, H.-J. Jeong, P. Antunes, A. Podobas, S. Raptis, V. Toy-Edens, M. Jin, S. Poedts, J. Magdalenic, G. Miloshevich

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